Shot Timer Mk1 Hardware Build Guide

Huckleberry Innovations · For firmware v1.7.5 · Oct 18, 2026 · Draft

This guide covers building a Shot Timer Mk1 from parts: the parts list, the protoboard layout, soldering in order, the checkout test, final assembly and packaging. Operating instructions are in the user manual; this guide is for building and servicing.

> **Draft.** This guide was put together from the original build notes and photos. Items marked ❓ still need to be checked against a real unit. Photos are from an earlier build (firmware v1.4.5, before the INA219 battery sensor was added), so a few wires differ from the tables, and the INA219 step has no photo yet. Where a photo and a table disagree, follow the table.

1. Parts and tools

Ref	Part	Qty	Notes
U1	Arduino Nano (ATmega328P) USB-C	1	Mounted on protoboard rows 16-30
U2	0.96in SSD1306 OLED 128×64 yellow/blue I2C	1	I2C 0x3C; mounted on underside of protoboard rows 5-8
U3	INA219 DC current sensor breakout (R100 0.1 ohm shunt)	1	I2C 0x40; added 10/16/2026 for battery gauge
PCB1	ElectroCookie solderable protoboard	1	
MK1	Clip-on lavalier microphone 3.5mm TRRS	1	Price was 9.99; check stock
J1	TRRS 3.5mm breakout board	1	Mic jack
BT1	9V USB-C rechargeable lithium battery	1	Rechargeable lithium only
BT1-CLIP	9V battery clip with leads	1	
SW1	Latching power switch blue-lit guarded	1	
SW2	16mm momentary push button green (START)	1	D9
SW3	16mm momentary push button red (STOP)	1	D8
LS1	Buzzer	1	D3
D1	White LED panel-mount	1	D13
CONN	2-pin wire connector/plug	3	Comes in pairs
R1	10k 1% resistor	1	Battery divider top (9V to A1)

R2a	2.2k 1% resistor	1	Battery divider bottom part 1 (2.2k + 4.7k = 6.9k)
R2b	4.7k 1% resistor	1	Battery divider bottom part 2
R3	100k 1% resistor	1	Mic bias divider top
R4a	22k 1% resistor	1	Mic bias divider bottom part 1 (22k + 68k = 90k)
R4b	68k 1% resistor	1	Mic bias divider bottom part 2
R5	2.2k 1% resistor	1	Mic supply resistor
C1	10uF electrolytic capacitor	1	Mic coupling cap
ENC	3D-printed PLA+ enclosure (top + bottom)	1	Printed in house
HW1	M2×4 socket head screw	22	
HW2	M2 flat washer	22	
CLIP	Belt clip	1	

Tools and supplies

- Soldering iron, solder, flush cutters, wire strippers, helping hands or a board holder
- Hook-up wire in red, black, yellow, blue, white and green; heat-shrink tubing and a heat gun
- Multimeter for continuity and voltage checks
- Kapton tape and double-sided foam tape (for the INA219)
- 1.5 mm hex key for the M2 socket-head screws
- Computer with the Arduino IDE and a data USB-C cable

2. How the protoboard is laid out

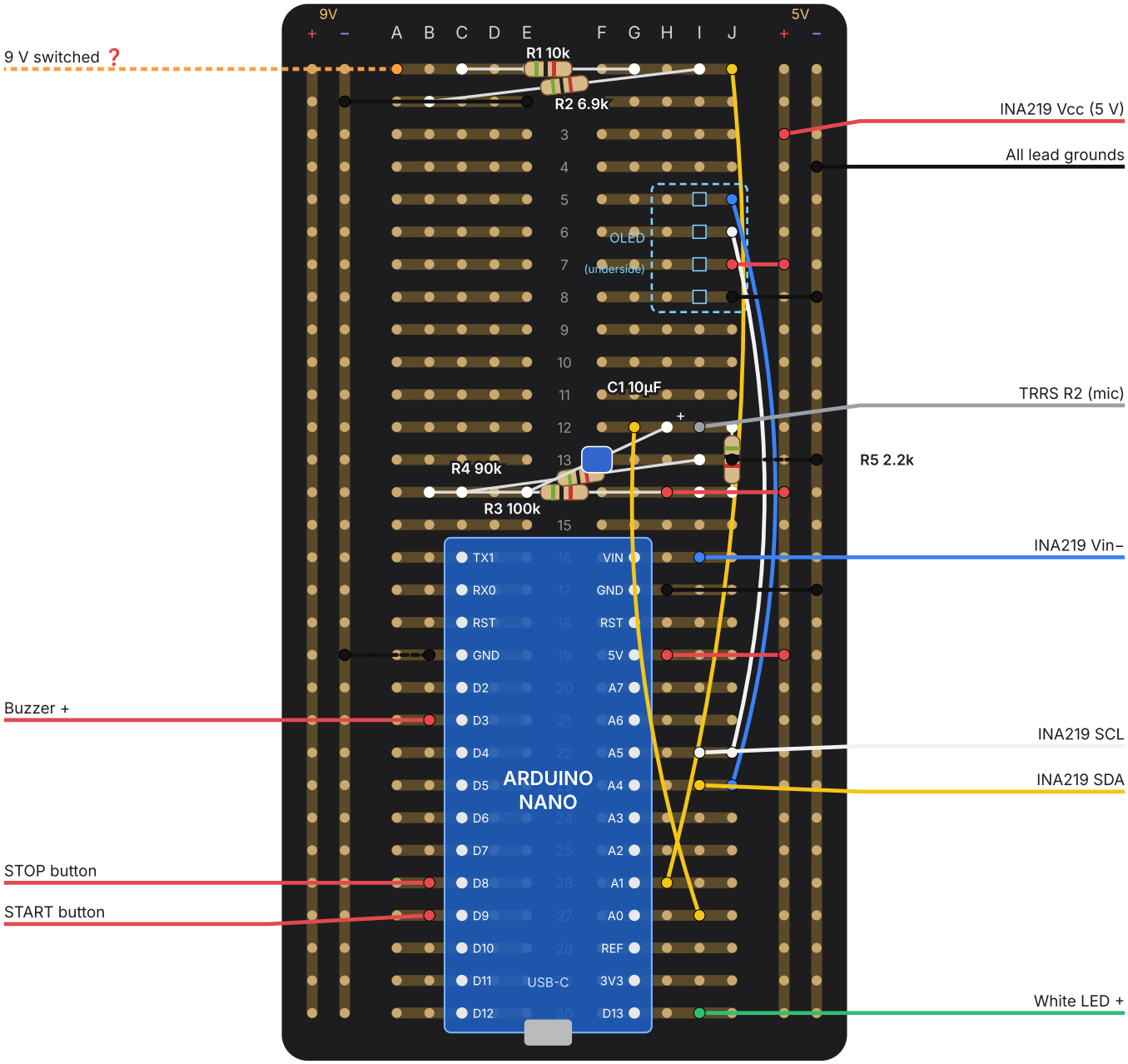
The board is an ElectroCookie solderable protoboard with 30 rows. Each row has two strips of five connected holes: **A–E** on the left and **F–J** on the right, split by the center gap. A part soldered into any hole of a strip is connected to everything else in that strip. Along each edge are two power rails: the left pair carries **9 V switched and GND**, the right pair **5 V and GND**.

The Nano straddles the center gap in columns C and G, rows 16–30. Its pins land in this order:

Row	Column C	Column G	Row	Column C	Column G
16	TX1	**VIN**	24	D6	A3
17	RX0	**GND**	25	D7	A2
18	RST	RST	26	**D8** STOP	**A1** battery
19	**GND**	**5V**	27	**D9** START	**A0** mic
20	D2	A7	28	D10	AREF
21	**D3** buzzer	A6	29	D11	3V3
22	D4	**A5** SCL	30	D12	**D13** LED
23	D5	**A4** SDA			

3. Board layout

Top view. Off-board leads leave the board on the side nearest their hole.



Dashed orange/black = to confirm on a real unit (❓). Strips: holes A–E in a row are connected; F–J in a row are connected.

4. Wiring by circuit

Power

From	To	Notes
Battery + (clip red)	Power switch input	
Power switch output (9 V switched)	Row 1 A–E (divider top) and INA219 Vin+	❓ **exact holes**
INA219 Vin−	Nano VIN (I16)	No direct switch-to-VIN wire may remain
Battery − (clip black)	GND	
Nano 5V (H19) and H14	Right + rail	
Nano GND (H17), E2	Right and left − rails	Ground paths joined at the board end

Battery divider (A1)

Part	Holes	Connects
R1 10k	C1 → G1	9 V switched → A1 node
R2 6.9k (2.2k + 4.7k)	I1 → B2	A1 node → GND
Jumper	J1 → H26	A1 node → Nano A1

Microphone (A0)

Part	Holes	Connects
TRRS R2 (signal)	I12	Mic node (row 12 F–J)
TRRS Sleeve	GND	
R5 2.2k	J14 → J12	5 V → mic node
C1 10 µF	+ H12, − E14	Mic node → bias node (row 14 A–E)
R3 100k	I14 → B14	5 V → bias node
R4 90k (22k + 68k)	C14 → I13	Bias node → row 13 F–J ❓ **GND**
Jumper	G12 → I27	Mic node → Nano A0

> ❓ On the board, A0 and the capacitor's + side are both on the mic node (row 12), while the KiCad sheet draws the 100k/90k divider biasing the mic. Confirm which matches the build before the schematic is redrawn.

I2C (OLED and INA219)

Signal	OLED	INA219	Nano
SDA	I5 → J5 → J23	Yellow → I23	A4
SCL	I6 → J6 → J22	White → I22	A5
Power	I7 → J7 → 5 V rail	Vcc → 5 V rail	5V
GND	I8 → J8 → GND rail	GND → GND rail	GND

Buttons, buzzer and LED

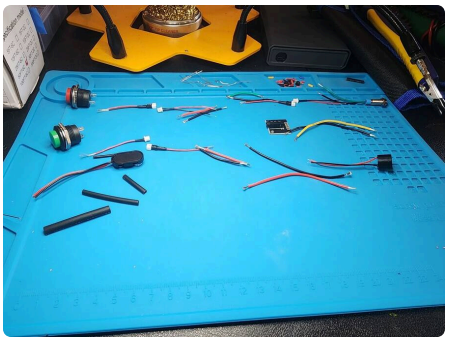
Part	Signal lead	Other lead
STOP (red 16 mm)	B26 → D8	GND
START (green 16 mm)	B27 → D9	GND
Buzzer	Red, B21 → D3	Black, GND
White LED	Green, I30 → D13	Black, GND ❓ **resistor**

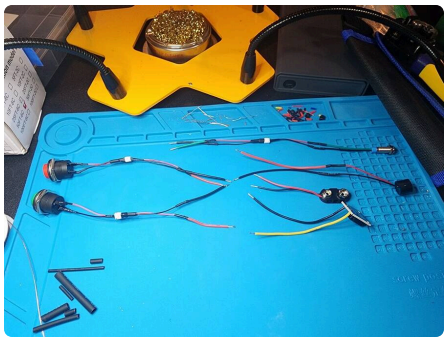
5. Soldering, step by step

Follow this order. It keeps the board accessible until the end, with the OLED going on last.

Step 1. Make the interconnect cabling

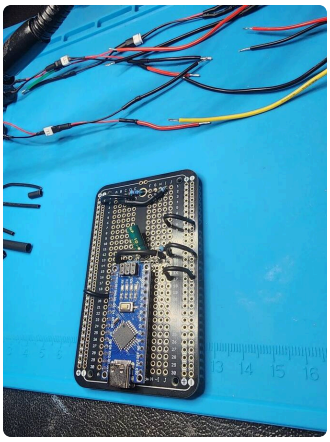
- Cut and strip lead pairs for every off-board part: buzzer, START, STOP, white LED, power switch, battery clip and TRRS breakout. Leads leave the board about 4 cm from its edge; the buzzer lead is about 9 cm.
- Fit the 2-pin connectors (3 pairs per timer) so the panel parts can be unplugged from the board for service.
- Slide heat-shrink over every joint before soldering, then shrink it.

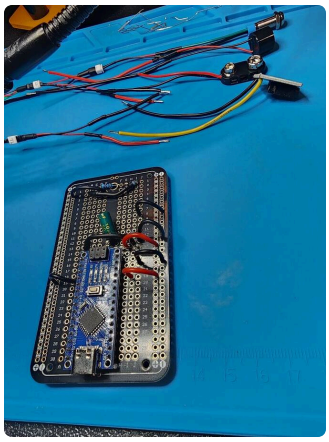




Step 2. Install the Nano, resistors, capacitor, 5 V and grounds

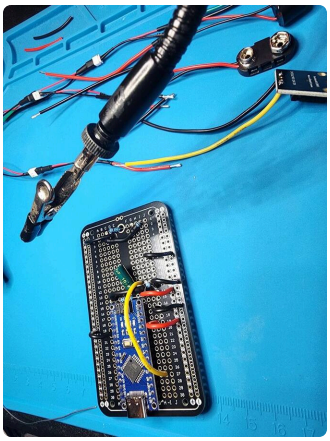
- Solder the Nano across the center gap in columns C and G, rows 16–30, with VIN at G16 and the USB-C end at row 30.
- Battery divider: R1 10k from C1 to G1; R2 (2.2k + 4.7k in series = 6.9k) from I1 to B2.
- Mic bias: R3 100k from I14 to B14; R4 (22k + 68k in series = 90k) from C14 to I13; R5 2.2k from J14 to J12.
- C1 10 µF: + leg in H12, – leg in E14. Check polarity before soldering.
- Run 5 V from H19 to the right + rail and H14 to the right + rail; GND from H17 to the right – rail and from E2 to the left – rail. Solder the ground paths together at the end of the board.

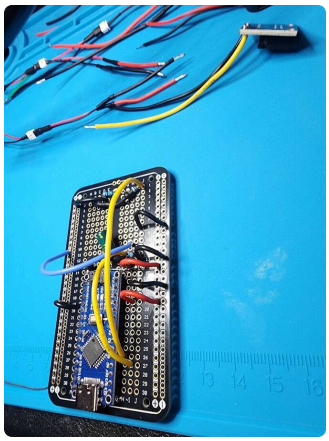




Step 3. Add the analog signal wires

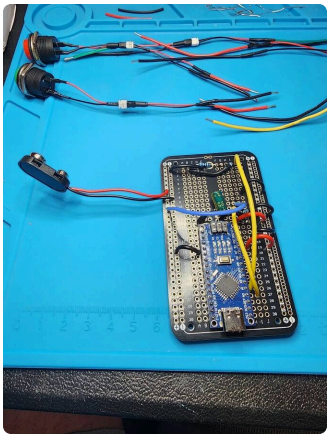
- A1 sense: jumper from J1 to H26.
- A0 mic: jumper from G12 to I27.





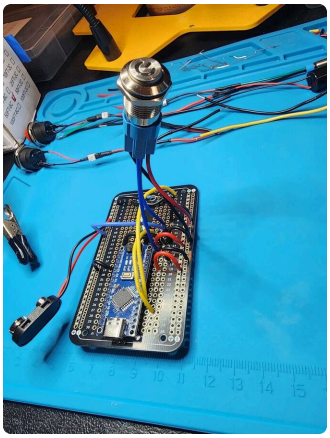
Step 4. Install the 9 V battery clip

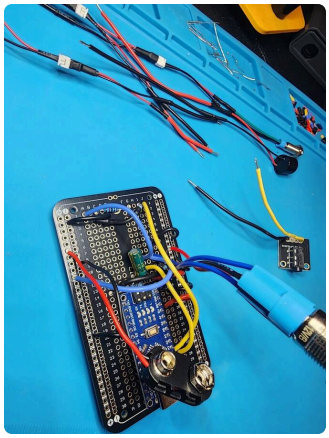
- Red lead to the power switch input (through its connector); black lead to GND.



Step 5. Install the power switch

- Switch input from the battery clip red lead. Switch output is the 9 V switched line: it feeds the divider top (row 1 A–E) and the INA219 Vin+ (next step). ❓ **Confirm the exact holes and wire colors.**
- Wire the switch's LED terminals so it glows blue when on. ❓ **Confirm the LED terminal connections.**
- Do **not** wire the switch output straight to Nano VIN any more; that path now goes through the INA219.



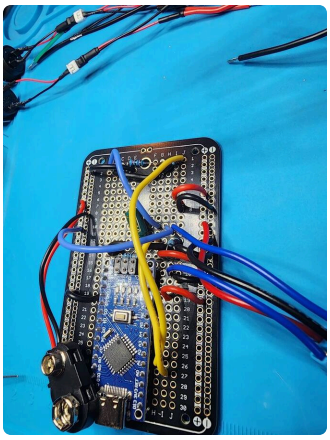


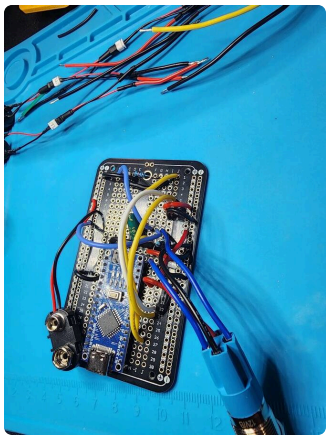
Step 6. Install the INA219 current sensor NEW 10/16/2026

- Cover the back of the INA219 with Kapton tape or heat-shrink so it can't short against the protoboard.
- Vin+ ← 9 V switched (switch output). Vin− → Nano VIN (I16, blue wire).
- Vcc → 5 V rail (never 9 V). GND → GND rail.
- SDA (yellow) → I23 (A4). SCL (white) → I22 (A5). Leave the A0/A1 address pads open (address 0x40).
- If a direct wire from the switch to VIN is still on the board, remove it; otherwise the shunt is bypassed and current reads about 0 mA.
- No photo yet. Add one from the next build.

Step 7. Install the OLED wires

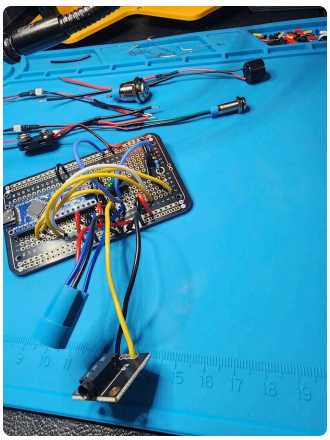
- The OLED header goes in I5–I8 from the underside: I5 SDA, I6 SCL, I7 VCC, I8 GND.
- Jumpers: J5 → J23 (A4, SDA), J6 → J22 (A5, SCL), J7 → 5 V rail, J8 → GND rail.
- Leave slack and route these wires where the enclosure won't pinch them. A pinched SDA wire leaves the screen dark.





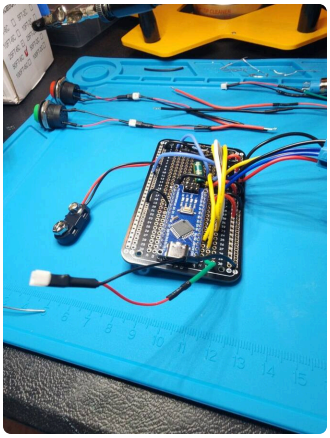
Step 8. Install the TRRS breakout

- R2 (mic signal) → I12. Sleeve → GND.



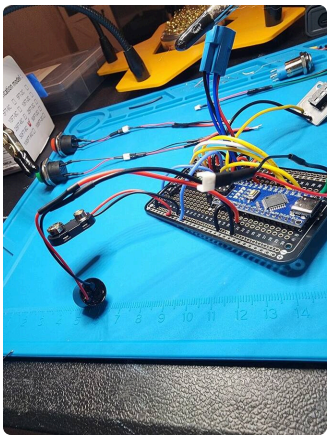
Step 9. Install the white LED cabling

- Green lead → I30 (D13). Black lead → GND. ❓ **No series resistor is on the board; confirm the panel LED has one built in.**



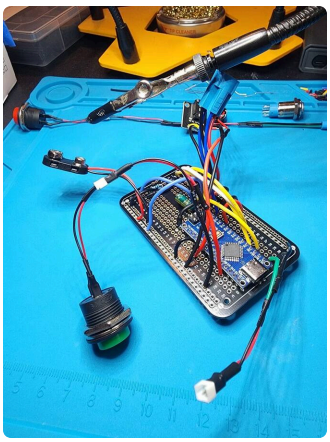
Step 10. Install the buzzer cabling

- Red lead → B21 (D3). Black lead → GND.



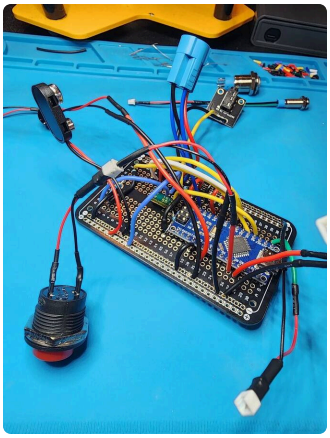
Step 11. Install the START button cabling

- Signal lead → B27 (D9). Other lead → GND.



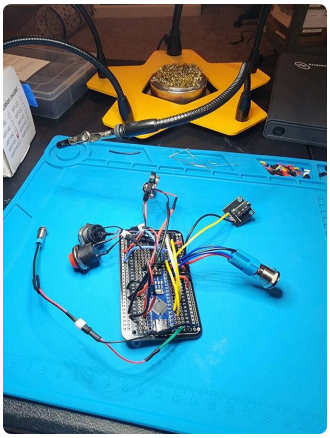
Step 12. Install the STOP button cabling

- Signal lead → B26 (D8). Other lead → GND.



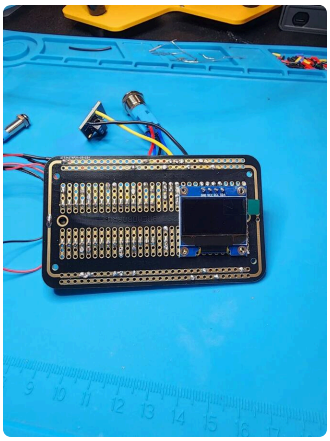
Step 13. Checkpoint: everything except the OLED

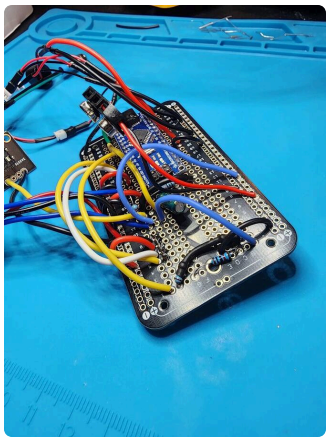
- Inspect every joint for bridges, especially around the Nano and rows 12–14.
- With a multimeter, check there is no short between 5 V and GND or between 9 V switched and GND.

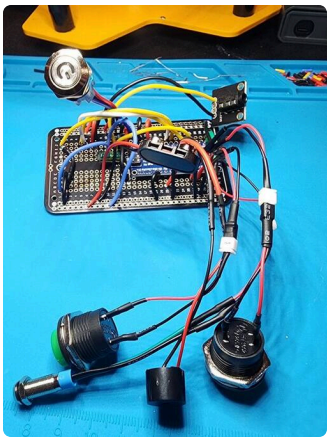


Step 14. Install the OLED last

- Solder the OLED onto its header on the underside of the board, screen facing out.
- Soldering it last keeps it out of the way while the wiring is done.

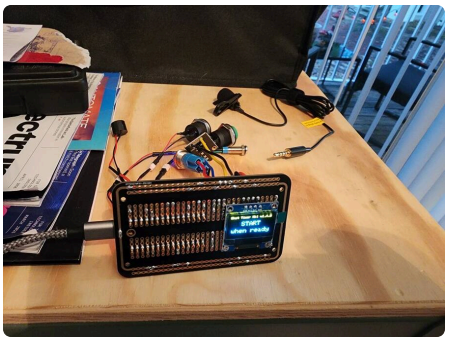


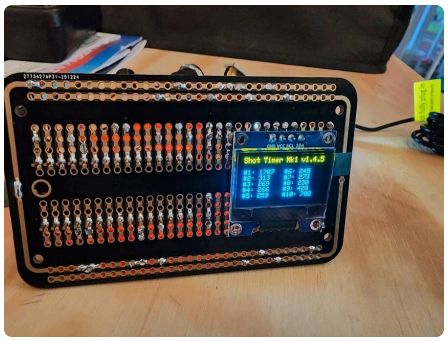




6. Checkout test

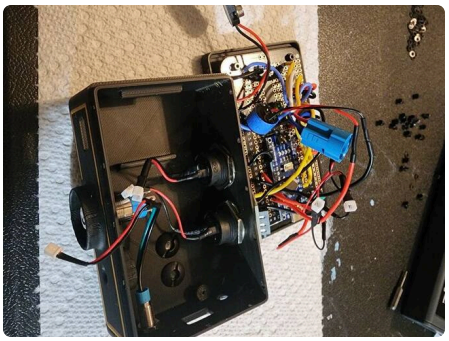
1. **Before power:** with the switch off, check with a multimeter that 5 V–GND and 9 V switched–GND are not shorted.
2. **Flash the firmware:** battery switch off, connect USB to a computer, upload `ShotTimerMk1_v1_7_5.ino`. A chirp from the buzzer on connect is normal.
3. **Start screen:** shows **Shot Timer Mk1**, **v1.7.5** and the battery gauge.
4. **Test mode:** hold START 2 s. Check **BATT:INA219** and RAM about 370. Clap near the mic: PEAK should jump. Tap STOP to exit.
5. **Battery reading:** unplug USB, switch the battery on, go to the BATTERY page. Expect about 9 V and a **positive** current of about 41 mA. A negative reading means Vin+ and Vin− are swapped; about 0 mA means the shunt is bypassed.
6. **Shot test:** tap START, wait for the beep (white LED on during the wait, blinking during the string), clap ten times, tap STOP. SPLITS should list 10 shots at the default threshold of 150.
7. **Buttons:** hold STOP 2 s to open the sound threshold screen, tap STOP to leave.
8. **Battery gauge:** charge the battery until its light turns green, then mark it FULL on the BATTERY page (hold START 3 s, tap STOP).
9. **Before shipping:** factory reset (hold START and STOP together 3 s) so the customer starts with no saved string and the default threshold. The learned battery capacity is kept.





7. Final assembly

- Mount the buttons, power switch, white LED and TRRS jack in the enclosure and plug their leads into the board connectors.
- Insulate the INA219 and fix it with foam tape where it can't touch the protoboard.
- Seat the board with the OLED lined up behind the screen window and fasten it with M2×4 screws and M2 washers.
- Route the I2C wires along the walls with slack so closing the lid doesn't pinch them.
- Fit the battery on its clip, close the enclosure and fasten the lid.
- Install the belt clip on the back of the enclosure.
- Switch on and confirm the start screen and battery gauge before packing.










8. Packaging and shipping

1. Cut the foam insert and set the Shot Timer Mk1 and microphone inside the box.
2. Add the printed quick-start card (and the user manual if included).
3. Add the Huckleberry Innovations label to the box.
4. If shipping: wrap the box in bubble wrap and place it in the shipping box.
5. Buy and print the shipping label from Pirate Ship, add the lithium battery caution label, and drop it at a USPS drop box. Follow current USPS rules for lithium batteries installed in equipment.








9. Open items

Item	To check
Power switch wiring	Exact board holes for the switch input, output and LED, and for INA219 Vin+.
Mic bias circuit	Whether A0 sits on the mic node (as on the board) or on the bias divider (as on the KiCad sheet).
Row 13 F–J	Confirm it is tied to GND (bottom of R4 90k).
White LED	Confirm the panel LED has a built-in resistor; the user manual says "D13 through 1k".
Ground wire at B19	Confirm where the GND wire from row 19 A–E lands.
Photos	Retake steps 5–6 with the INA219 installed on the next build.

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